Which jet observable tags quarks and gluons best at the EIC?

How the ROC comparison works, and what it told us

Photoproduction dijets, $\sqrt{s} = 64$–300 GeV

The question

Gluon jets are wider than quark jets. Everybody knows this.

But "wider" is not an observable. You have to *choose a number* that measures it — and there are dozens of candidates.

Nobody had checked which one actually works at EIC energies.

So we took three, computed them on the *same jets*, and raced them.

That last bit matters more than it sounds. Same clustering, same cuts, same event loop, all four energies. If one wins, it is the observable that won — not the sample, not the reconstruction.

The three contestants

Integrated jet shape

$\Psi(r = 0.3)$

What fraction of the jet's energy sits within r of the axis?

Quark jets are collimated, so their Ψ is *larger*.

No algorithm. No threshold. Just distances and energies.

Subjet multiplicity

n_{subjets}

Recluster with k_T, stop at a resolution scale, count the lumps left.

$y_{\text{cut}} = 5 \times 10^{-4}$, the old HERA convention.

Remember this one. Its threshold has a surprise in it.

Soft-drop multiplicity

n_{SD}

Walk the jet's splitting tree, count splittings where the softer prong carries $z > 0.1$.

Theory's favourite. Approaches C_A/C_F asymptotically.

Each needs a *different* reclustering, because each asks a different question: n_{subjets} needs a momentum scale (k_T), n_{SD} needs angle ordering (Cambridge/Aachen), Ψ needs nothing at all.

Section IV: the four formulas

1. The two efficiencies. Pick a pass mark c. Ask what fraction of each class survives:

$$\varepsilon_{\mathrm{GG}}(c) = \Pr(X \geq c \mid \mathrm{GG}), \qquad \varepsilon_{\mathrm{QQ}}(c) = \Pr(X \geq c \mid \mathrm{QQ})$$

One you want high (gluons kept). One you want low (quarks sneaking in). You can never have both.

2. The area under the curve.

$$\mathrm{AUC} = \int_0^1 \varepsilon_{\mathrm{GG}} \, \mathrm{d}\varepsilon_{\mathrm{QQ}}$$

The probability that a random gluon jet outscores a random quark jet. 0.5 = coin flip, 1 = perfect.

3. Mean separation. $\mathcal{S} = \dfrac{\langle X \rangle_{\mathrm{GG}} - \langle X \rangle_{\mathrm{QQ}}}{\sqrt{\sigma^2_{\mathrm{QQ}} + \sigma^2_{\mathrm{GG}}}}$ gap between the means, in units of their width.

4. Purity. $\mathrm{Purity}_{\mathrm{GG}} = \dfrac{N_{\mathrm{GG}}\varepsilon_{\mathrm{GG}}}{N_{\mathrm{QQ}}\varepsilon_{\mathrm{QQ}} + N_{\mathrm{GG}}\varepsilon_{\mathrm{GG}} + N_{\mathrm{GQ}}\varepsilon_{\mathrm{GQ}}}$

What an experimentalist actually gets: of the jets that pass, how many are really gluons?

What "c" means — a worked example

The rule is always the same shape: **call it a gluon jet if $X \geq c$.**

c is a *pass mark*. Like a pass mark on an exam — a line you draw, not a fact about the students.

Set $c = 5$ for n_{SD} and count what happens to real HERA jets:

	called GLUON	called QUARK	total
really a gluon	14 524	17 466	31 990
really a quark	20 574	75 431	96 005

Read it *by row*, because each row is a class whose total you know:

$$\varepsilon_{\mathrm{GG}} = \frac{14\,524}{31\,990} = 0.45 \qquad \varepsilon_{\mathrm{QQ}} = \frac{20\,574}{96\,005} = 0.21$$

That is Formula 1. You caught 45% of the gluons; 21% of the quarks fooled you. The vertical bar "$|\,\mathrm{GG}$" is just telling you to divide by the row total.

How the ROC plot is built

Now try *every* pass mark, not just one:

c	$\varepsilon_{\mathrm{QQ}}$	$\varepsilon_{\mathrm{GG}}$
3	0.94	0.99
4	0.73	0.95
5	0.41	0.78
6	0.15	0.45
7	0.03	0.15

Each row is **one dot** on the plot.

Join the dots $\rightarrow$ the ROC curve.

Area underneath $\rightarrow$ the AUC.

For n_{SD}, c takes twelve values:

$$c = 0, 1, 2, \ldots, 11$$

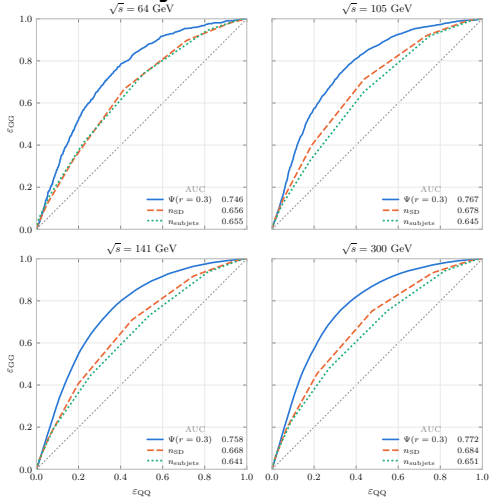
Only whole numbers — a jet has 4 splittings or 5, never 4.5.

For Ψ, 400 values between 0 and 1.

That is why Ψ's curve looks smooth in our figures and the counting curves look faceted. *The faceting is honest*: there really are only twelve places you could put that cut.

Why sweep at all? If we quoted one cut, you could fairly say we picked the one that flatters our favourite. Sweeping removes the choice entirely.

Result 1: the jet shape wins, everywhere



$\sqrt{s}$ [GeV]	64	105	141	300
$\Psi(0.3)$	0.746	0.767	0.758	**0.772**
n_{SD}	0.656	0.678	0.668	0.684
n_{subjets}	0.655	0.645	0.641	0.651

Why the simplest observable wins

Both counting observables boil the jet down to *how many splittings clear a threshold*. Everything below the threshold is thrown away.

At EIC energies a 10–20 GeV jet has only **three or four** splittings that qualify. You are counting on a very coarse ruler.

Our own data says so. Theory predicts the gluon/quark ratio of n_{SD} should approach $C_A/C_F = 2.25$. We measure:

$$\frac{\langle n_{\mathrm{SD}} \rangle_{\mathrm{GG}}}{\langle n_{\mathrm{SD}} \rangle_{\mathrm{QQ}}} = \frac{4.41}{3.47} = 1.27$$

Barely half the asymptotic value. The counting argument is *not sharp* here.

$\Psi(r)$ throws nothing away — every constituent contributes, weighted by its energy.

When there is little to count, don't count.

Result 2: $\Psi(0.3) < 0.6$ was a good choice

The companion paper picked its thin/thick cut by eye, years ago. Was that lucky or right?

cut on $\Psi(0.3)$		gluons kept	quarks let in
< 0.60	(the paper's choice)	0.712	0.289
< 0.63	(the optimum)	0.758	0.329

Sweeping every possible cut and asking which maximises $\varepsilon_{\mathrm{GG}} - \varepsilon_{\mathrm{QQ}}$ gives **0.63**.

The paper chose **0.60**.

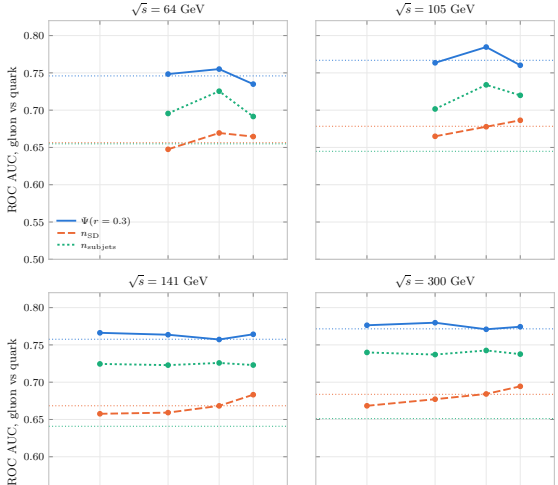
Essentially optimal. Not a lucky guess — the right call.

The other cut, $\Psi > 0.8$ for quark-like jets, reads in reverse: it keeps 37% of quark jets while admitting only 6% of gluon jets. A small but very clean quark sample.

Result 3: the counting observables swap places

Pooled over η: n_{SD} beats n_{subjets}. That is what the literature predicts.

Binned in η: n_{subjets} wins — in **all fourteen bins**, at every energy, no exceptions, by 0.027 to 0.072.



An analogy for what is going wrong

Imagine ranking students by raw exam score — but different schools set papers of different difficulty.

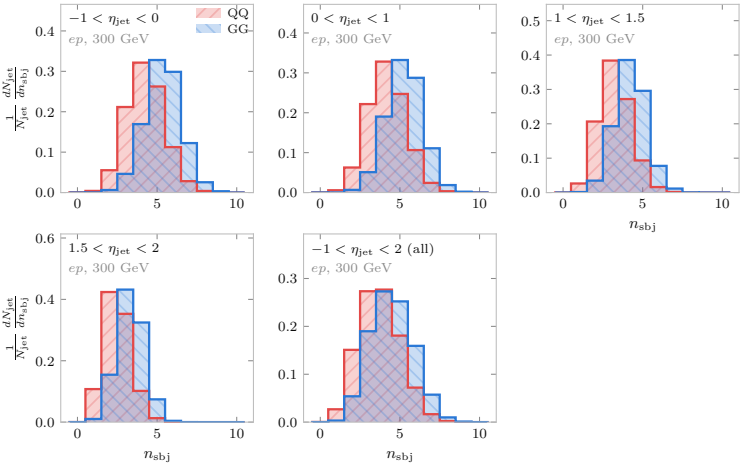
- **Within one school**, the ranking works fine.
- **Pool all schools**, and you mostly measure *which school they attended*, not who is better.

That is exactly what happens to n_{subjets} across η.

Its mean slides from **4.25** in the most backward bin to **2.49** in the most forward one. A gluon jet out front looks like a quark jet in the middle.

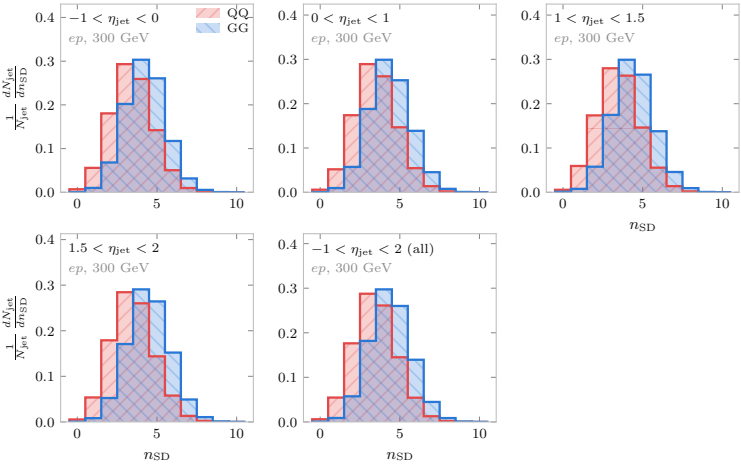
Splitting the pooled variance: **27–39%** of the width of n_{subjets} is this sliding, not class information. For n_{SD} it is **0.2%**.

See it directly



$n_{\rm subjets}$: the whole distribution marches *left* as η grows — peak at 5, then 5, then 4, then 3. The last panel pools them all: broader, flatter, far more overlap. The QQ–GG shift drops to 0.75, *smaller than in any individual bin*.

The same plot for n_{SD}



n_{SD}: nothing moves. Every panel looks the same, and the pooled panel looks like the others — because there is nothing to smear.

Two observables, same jets. One has a problem the other does not.

The physics: a moving ruler

The usual story is that n_{subjets} counts above a *fixed momentum scale*. In the standard implementation it does not.

The y_{cut} is normalised to the summed constituent **energy**:

$$\min(p_{T,i}, p_{T,j})\,\Delta R_{ij} \;>\; \sqrt{y_{\mathrm{cut}}}\; E_{\mathrm{jet}}$$

Write $p_T = z\,p_T^{\mathrm{jet}}$ and $E = p_T^{\mathrm{jet}} \cosh\eta$. The jet p_T **cancels**, leaving

$$\boxed{z\,\Delta R \;>\; \sqrt{y_{\mathrm{cut}}}\;\cosh\eta}$$

So the effective threshold *grows as you go forward*:

η	0.0	1.5	2.5
z_{eff}	0.022	0.053	**0.137**

while n_{SD} cuts at $z > 0.1$ *everywhere*.

Three ways we checked it

1. Energy, not p_T, is what it responds to.
Binned in jet energy: $\langle n_{\text{subjets}} \rangle$ falls 4.23 → 1.50. $\langle n_{\text{SD}} \rangle$ holds at 3.43–3.50.

2. The functional form is exactly right.
Counting grows like $\ln(1/z)$, so the count should fall linearly in $\ln\cosh\eta$. Measured at fixed p_T:

$$\langle n_{\text{subjets}} \rangle = 4.35 - 1.53 \ln\cosh\eta, \qquad r^2 = 0.991$$

3. It predicted a crossover we had not measured.
z_{eff} passes 0.1 near $\eta \approx 2.2$, so beyond that n_{SD} should retake the lead. Extending the bins:

η bin	[1.5, 2)	[2.5, 3)	[3, 4)
n_{SD}	0.694	**0.694**	**0.698**
n_{subjets}	**0.738**	0.682	0.550

Predicted 2.2. Observed between 2.25 and 2.75.

So which counter is actually better?

n_{subjets} is — where its threshold is soft.

Centrally it resolves emissions about **five times softer** than n_{SD} ($z \approx 0.022$ against 0.1), so it samples far more of the Casimir-weighted radiation. More information per jet.

Its weakness is not the counting. It is the **moving threshold**.

Pooled across η, the same observable is effectively a different measurement in each part of the range — so the classes smear together.

Practical upshot:

- Quoting *inclusively*? Use n_{SD}.
- Quoting *differentially in* η? Use n_{subjets}.
- Or fix it at source: scale y_{cut} by $1/\cosh^2\eta$, or recluster in a frame where the jet is central.

What we recommend

For the first generation of EIC jet-substructure measurements, use the integrated jet shape $\Psi(r)$.

- Best separation of the three, at every energy
- Flat across the whole EIC range, and matches the HERA reference
- **No internal tagging parameters** to defend — important at a facility with no data to calibrate against
- Directly comparable to archival ZEUS measurements, a validation path no newer observable has

And a concrete warning: do not quote n_{subjets} inclusively over a wide η range without saying so. Its discrimination power is real, but pooling throws much of it away.

Honest limits

- **Generator level only.** No detector effects. The comparison is relative, so smearing should hit all three similarly — but we have not shown that.
- **Three observables, not the landscape.** Angularities, energy-flow polynomials and energy correlators are not here.
- **No statistical uncertainties quoted on the AUCs.** Two jets from one event are correlated, so honest error bars need bootstrapping over *events*, not jets.
- **The $1/\cosh^2\eta$ fix is suggested, not tested.**

What does not change: $\Psi(r = 0.3)$ wins at every energy, in every η bin, at every bin width.

One-slide summary

1. A ROC curve is built by sliding a **pass mark** through every possible value and recording two numbers each time: gluons kept, quarks let in.
2. The **AUC** is the area underneath — the chance a random gluon jet outscores a random quark jet. It contains no choice of cut.
3. $\Psi(0.3)$ wins at every energy (0.75–0.77), because counting observables have too little to count at 10–20 GeV.
4. The companion paper's cut at **0.6** is essentially the optimum (0.63). Good call.
5. n_{subjets} and n_{SD} swap places depending on whether you pool or bin in η — because the k_T threshold is normalised to *energy*, so it tightens as $\cosh\eta$.